Collected Mathematical Papers

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Papers 665 (conclusion), 666, 667, 668, 669, 670 (commencement)

Arthur Cayley

Contents

1 [665] A Memoir on the Double ϑ-Functions (conclusion)

[From the Philosophical Transactions of the Royal Society of London*, vol. 169 (1878), pp. 663–730; continued from preceding pages.]*

(Page 201.) We have, as the equation of a quadric surface enveloped by

$$\alpha\,(x-x')+\beta\,(y-y')+\gamma\,(z-z')+\delta\,(w-w')=0,$$

which is the same form as before, the relations

$$ab'+ba'=0,\quad cd'+dc'=0,\quad \text{etc.},$$

or, more precisely,

$$\frac{\partial\Omega}{\partial x}\cdot X'+\frac{\partial\Omega}{\partial y}\cdot Y'+\frac{\partial\Omega}{\partial z}\cdot Z'+\frac{\partial\Omega}{\partial w}\cdot W'=0.$$

It is to be remarked that there are 15 expressions such as $A1B+B1A$, and 45 expressions such as $AC1BD+AD1BC$; and that each of these $(15+45=)60$ expressions is a quadric function of ∂_x, ∂_y, ∂_z, ∂_w alone.

In like manner Ω, Ω_1, Ω_2, Ω_3 are quadric functions of (x,y,z,w), and similarly Ω', Ω_1', Ω_2', Ω_3' of (x',y',z',w'); and hence Ω, Ω_1, Ω_2, Ω_3 satisfy the conditions

$$\frac{\partial\Omega}{\partial x}-2x+\frac{\partial\Omega_1}{\partial y}\cdot 2+\frac{\partial\Omega_2}{\partial z}\cdot 2+\frac{\partial\Omega_3}{\partial w}\cdot 2=0,$$

and similarly for the conjugate functions.

Expression of $\Theta\,\Omega-\Omega\Theta'$*: second subheading.*

We assume $\Theta\,\Omega-\Omega\Theta'=\frac{1}{2}M\,\Omega''$, where M is a quadric function of ∂_x, ∂_y; suppose

$$M\equiv\Theta\,(xx'+yy')+\Theta'\,(xx'+yy').$$

It is to be remarked that there are 15 expressions which can be presented in the form

$$\Big(\Theta x'\frac{\partial}{\partial x}+\Theta y'\frac{\partial}{\partial y}\Big)\sqrt{X}=\Big(\Theta x'+\Theta y'\cdot\frac{xy'}{X}\Big)\sqrt{X}+\Big(\Theta y'\cdot\frac{xx'}{Y}\Big)\sqrt{X};$$

it is clear that the term in question is at once expressible in the form indicated by means of these.

It is to be remarked that there are 15 expressions which form a quadric system of (x,y) alone; viz., X, Y, $\frac{xy'}{X}$ etc. are linear functions of (x,y) together with the symbols

$$\Big(\frac{\partial\Theta}{\partial x}+\frac{\partial\Theta}{\partial y}\Big)\sqrt{X}+\Big(\frac{\partial\Theta}{\partial x'}+\frac{\partial\Theta}{\partial y'}\Big)\sqrt{Y},$$

where the bracketed symbols satisfy the conditions

$$\xi\alpha+\eta\beta=0,\qquad \nu\alpha+\mu\beta=0,$$

and the like for ξ', η', ν', μ'.

Putting for the moment

$$\lambda\alpha+\mu\beta=\nu\alpha,\qquad \mu\beta+\nu\gamma=\mu\beta,\qquad \nu\gamma+\lambda\delta=\nu\gamma,\qquad p\alpha+q\beta=r\alpha+s\beta,$$

$\mu=ab$, $\nu=ac$, ...; the values of ξ, η, ν, μ, ξ', η', ν', μ' are

$$\xi=0,\qquad \eta=0,\qquad \nu=0,\qquad \mu=0,$$

respectively.

I found it convenient to assume

$$\Theta = -2\big(x'^2 + y'^2\big)E = b\,xx' - ay\,y' + p\,(yy' + xx') + q\,(yy' - xx'),$$

and that the system of seven is

$$\Theta = -2\,(x'^2 + y'^2)E - \alpha\,xy' + \beta\,yx' + (\gamma + \delta)\,xx' + (\gamma - \delta)\,yy'.$$

(Page 202.) Assuming now

$$\Theta = -2\,(x'^2 + y'^2)\varphi + (\alpha + \beta)(x\,xy' - y\,xy)$$

where Θ has to be determined so that the first of the same conditions may be satisfied; then substituting this value of Θ, we have

$$\Big(\Theta y' - py' + qy' - ry' - s\,y'\Big)\sqrt{X} = \Big(\Theta x' - px' + qx' - rx' - s\,x'\Big)\sqrt{Y};$$

that is,

$$-a\,h\,x\,y' - x\,b\,x'\,y\sqrt{X} = \Big(- ax\,x' + b\,xy' - b\,y\,x'\Big)\sqrt{Y};$$

viz. in the inner subscription of $\sqrt{X}$, the function Θ is to be diminished by

$$(a\,bc + b\,ca)\sqrt{acbcdabcd} = (acbcdabcd)\sqrt{abcdabcdc},$$

or, what is the same thing,

$$\Big(- h\,b\,xx' - h\,a\,xx'\Big)\sqrt{X} = (-yh\,b\,xx' - h\,a\,y\,xx')\sqrt{Y};$$

also,

$$\sqrt{Yx} = \tfrac{1}{2}a(abcdefxy)\sqrt{xy},$$

and we thus reduce the equation to

$$-(\sqrt{abcdef}\cdot\sqrt{abcdefc})\sqrt{xy}\cdot\big(abcd\cdot\sqrt{abcdef}\big).$$

But, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$\frac{d}{dx}\sqrt{ab} = -\big(\sqrt{abcd}\cdot a_1 - \sqrt{abcd}\cdot a_2\big)\sqrt{ab};$$

and the formula thus becomes

$$-(\sqrt{abcdef} - \sqrt{abcdefc})\cdot\frac{d}{dx}\sqrt{ab};$$

and, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$2\sqrt{ab}\cdot\frac{d}{dx}\sqrt{ab} = \frac{d}{dx}\cdot a\,b;$$

consequently

$$\Theta\alpha = -(\sqrt{abcd} - \sqrt{abcdc})\frac{1}{2}\,\frac{d}{dx}\cdot a\,b,$$

and the value of Π thus, suppose

$$\Pi = \tfrac{1}{2}\,(\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1)\cdot a\,b,$$

or so they may be written,

$$\Pi = \tfrac{1}{2}\,(\alpha b\,c\, - a\,b_1c_1)(b\,c\,d - a_1c_1d_1) + a^2 - 2axy\,y + a^2c_1\sqrt{X\,Y}.$$

(Page 203.) Hence

$$abc - a_1b_1c_1 = (a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) - (a + a_1)(b + b_1)^2 - 5c_1^2;$$

and similarly,

$$def - d_1e_1f_1 = (d + e - d_1 + e_1 + f_1)(e + f - d_1 + e_1 + f_1) - (d + d_1)(e + e_1)^2 - 5f_1^2.$$

The expression for $a^2c_1^2$ contains therefore the terms

$$-(a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) + (a + a_1)(b + b_1)^2 + 5c_1^2,$$

or that, $A = abcc_1$, where A is something like the coefficient of $\beta\gamma$ in

$$(\alpha + \beta + \gamma + \delta - \epsilon - \eta)\sqrt{XY},$$

which, viz. in fact, the sum of Camy's form in ϕ, equation (200), his ϕ being

$$= 95'' - a.$$

It is to be observed that, Π is the difference of two squares of expressions

$$A\Pi'' B\Pi'',$$

the radical which multiplies the being

$$\frac{1}{a + b\,\cos T + c\,\cos T'\sqrt{ab}} \cdot \sqrt{ab - a_1\cos T - b_1\cos T'}.$$

the differential becomes

$$\frac{dT\cdot\sqrt{ab - a_1\cos T - b_1\cos T'}}{(\cos^2 T\cdot\frac{a+b}{1+\alpha\,\cos T'} + \beta\,\sin^2 T)\,\sqrt{a + b\,\cos T + c\,\cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q\,\cos T + R\,\cos T'}\sqrt{a + b\,\cos T + c\,\cos T'}}(a+b\,\cos T+c\,\cos T')(1+\alpha\,\cos T+\beta\,\cos T').$$

The denominator could, of course, be reduced to the form $(\alpha 1, \cos T, \sin T)^2$; but the arrival form seems preferable, inasmuch as it puts in evidence the linear factors

$$\frac{1}{(a + b\,\cos T + c\,\cos T')\,\sqrt{a + b\,\cos T + c\,\cos T'}\sqrt{XY}}(\alpha + \beta\,\cos T + \gamma\,\cos T')\,dT;$$

and there seems to be no advantage in further reducing the integral.

(Page 204.) Assuming now

$$\Theta = -2x'\sqrt{a^2 + y^2} + (\alpha' + \beta')\sqrt{X}\cdot(\Theta x' - px' + qx') - rx'(\sqrt{X} - \sqrt{Y}),$$

where Θ has to be determined so that the first of the same conditions may be satisfied. Then, substituting this value of Θ in the equations

$$\Big(\Theta y' - py' + qy' - ry'\Big)\sqrt{X} = \Big(\Theta x' - px' + qx' - rx'\Big)\sqrt{Y},$$

that is,

$$\Big(-axy' - x\,by'x'\Big)\sqrt{X} = \Big(-axx' + bxy' - byx'\Big)\sqrt{Y},$$

viz. in the inner subscription of $\sqrt{X}$, the function Θ is to be diminished by

$$(axbc)\cdot\sqrt{abcdef\cdot\sqrt{XY}} - \sqrt{abcdef\cdot\sqrt{YX}} = -\tfrac{1}{2}\sqrt{X} - \tfrac{1}{2}\sqrt{Y};$$

or, what is the same thing,

$$(abc - a_1b_1c_1)\sqrt{X} = \sqrt{abcd\cdot X} - \sqrt{abcd\cdot Y},$$

that is,

$$abc - a_1b_1c_1 = \sqrt{abcd\cdot X/Y} - \sqrt{abcd\cdot Y/X}.$$

But, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$\frac{d}{dx}\sqrt{ab} = -\left(\sqrt{abcd}\cdot a_1 - \sqrt{abcd}\cdot a_2\right)\sqrt{ab},$$

and the formula thus becomes

$$-\left(\sqrt{abcdef} - \sqrt{abcdefc}\right)\cdot\frac{d}{dx}\sqrt{ab};$$

and, referring to the expression for $\lambda\sqrt{ab}$, we have, by a mere interchange of letters,

$$2\sqrt{ab}\cdot\frac{d}{dx}\sqrt{ab} = \frac{d}{dx}(ab);$$

consequently

$$\Theta\,a = -\left(\sqrt{abcd} - \sqrt{abcdc}\right)\cdot\tfrac{1}{2}\,\frac{d}{dx}(ab),$$

and the value of Π thus, suppose

$$\Pi = \tfrac{1}{2}\left(\sqrt{abcd} - \sqrt{abcdc}\right)(a_1 - b_1)\,ab,$$

or so they may be written,

$$\Pi = \tfrac{1}{2}\,(abc - a_1b_1c_1)(bcd - a_1c_1d_1) + a^2 - 2axy + a^2c_1\sqrt{XY}.$$

(Page 205.) *Expression of* $\Theta\,\Omega - \Omega\,\Theta'$: *second subheading.*

Writing for shortness $\Theta\,\Omega - \Omega\,\Theta' = R$, so before, and as in other cases ... in general $\Delta PQ = P\,\Delta Q + Q\,\Delta P$. Hence starting from $A = \Pi\,abc' - a'b'c$, we have

$$\Delta A = \Pi\,\Delta(abc') - \Pi(a'b'c)$$

$$= 4c'\left(\Pi\,\frac{PU}{QV} - \Pi\,\frac{PV}{QU}\right)(\Omega - \Omega')\,c'\xi\xi' + 2b\,(b - b')(\xi - \xi')(\Omega - \Omega'),$$

where

$$\Delta a = -\,a + (\xi - \xi')aa', \quad \xi - \xi' = 0, \quad \xi - \xi' = -1 \quad (\xi a - \xi'\xi),$$

so that

$$\Delta a = a - a + (\xi - \xi')aa' = \tfrac{1}{2}(a - a_1)(b + b_1)(\xi - \xi') = aa'(b - b_1).$$

We thus have

$$\Delta A = -\,\Pi abc'[(\xi - \xi')aa'c_1c_1' + \Pi bb'c'c_1'c(\xi - \xi')] - \tfrac{1}{2}(a + a_1)(b + b_1)(c + c_1).$$

But the expression for Π contains the factors $4c'$, a, b, c; hence multiplying the whole by A, we have

$$\frac{4}{\Pi}\Delta A = \Pi a^2[\Pi(bc' - b'c)(bc - b'c') + 4\alpha b\,(\xi - \xi')]$$
$$-\tfrac{1}{2}\Pi b'(ab - a_1b_1)(c - c_1) - \Pi ab\,(a + b)(\xi - \xi')(\sqrt{XY})$$
$$+\tfrac{1}{2}\Pi\big(abcdc' - (\sqrt{XY})\big) - \omega_1(\xi - \xi')(\xi - \xi'),$$

where M has its foregoing value $= [\Pi_1 - (\sqrt{XY})](\Pi' \cdot \Pi w + W(bw'))$.

First step of the reduction.

Writing $\Theta\xi' = U$, $b_2acdef = U_1$, then $X = aU$, $Y = a_1U_1$, and consequently

$$X - U = a'U^2, \qquad Y - U_1 = U'a_1U_1^2;$$

the second expression in U, U_1 denoting differentiations in x, y respectively; then

$$X' - X = -aX + 2\Theta x - \tfrac{1}{2}a'\big(\frac{\partial U}{\partial x}\big)^2 \cdot \sqrt{ab'/aac'},$$

and similarly

$$Y' - Y = -aY + 2\Theta y - \tfrac{1}{2}a'\big(\frac{\partial U}{\partial y}\big)^2 \cdot \sqrt{ab'/aac'},$$

where, as before, Θ has been treated in the analytic form expressed in the formulae.

(Page 206.) Hence

$$\Theta\,\Omega = a \cdot \Omega - \frac{2}{a}\Big(\Theta\,\frac{P\,V}{Q\,U} - \Theta\,\frac{P\,U}{Q\,V}\Big)(\Omega - \Omega')\,2\Theta(ba - b'a) - (aa_1)(b - b')(\xi - \xi');$$

where the first value, Ω, substituting for X, Y, Ω, Ω' their values, and arranging according to ∂x, ∂y, Ω, is

$$= \frac{2}{a}\Big[a'(\partial'_x P)(\partial'_x Q) - a'(\partial'_y Q)(\partial'_y P) - a'\,\partial'_x\partial'_y(\partial'_x\partial'_y)\Big] + \cdots$$
$$+2b'(aa_1 - a_1a)(xy' - x'y) + \cdots$$
$$+(\partial x' - \partial' x')ab'_1 + a(b_1b' - b\,b'_1)(\partial x' \cdot \partial'_x\partial'_y \cdot \partial'_y P \cdot \partial'_y Q) \cdot aa'.$$

After collecting and simplifying, one is led to expressions of the form

$$\Theta\xi' = \tfrac{1}{2}\big(\frac{\partial\Pi}{\partial x} \cdot a' - \frac{\partial\Pi}{\partial y} \cdot a\big),$$

and these formulae are no advantage in further reducing the integral.

(Page 207.) *Third step of the reduction.*

First, for the coefficient of $(\partial\omega')^2$, multiplying that $\Delta = a_1\,\omega - \omega_1$, we have

$$2\Big(\frac{\partial U}{\partial x}\Big)^2 a' - 2U(\alpha + a_1) - 2U'(\alpha' + a_1)(a + a_1)(\xi - \xi') - 2U'(a_1b_1 - ab)(\xi - \xi') - 2U'(b'_1a_1 - ba)(\xi - \xi')$$
$$-(\xi - \xi')(\xi - \xi')a = a(\partial x'^2 \cdot \partial y'^2 \cdot \partial x' \cdot \partial'_y \cdot a_1).$$

Adding these, the right-hand side divides by $(a_1 - a)$; that is, by $2P$; and the resulting value is

$$= \tfrac{1}{2}a'\xi\xi' aba'\sqrt{ab'abc'},$$

the term $aaa'ac\xi\xi'$ is reduced to the foregoing formulae; by performing the additions, observing as before, that

$$\Theta = a'b'\Theta\xi' a' = a'(a_1 b - 2a'b')(\xi - \xi') - 2aa_1b_1c_1\xi.$$

Adding these, the right-hand side divides by $(a_1 - a)$; that is, by $2P$; and the resulting value is

$$(\xi-\xi')(\Theta\xi')(a_1-b_1)(c-c_1)-2a+b_1+b'(\xi-\xi')+b'(\xi-\xi')(\Theta\xi')(a_1-b_1) = -2a+b_1+b'(\xi-\xi'),$$

or that, $A = abcc_1$, where A is something like the coefficient of $\beta\gamma$ in $(\alpha+\beta+\gamma+\delta-\epsilon-\eta)\sqrt{XY}$, in the manner above ξ ... in fact, the sum of Camy's form in ϕ equation (200), his ϕ being $= 95''-a$.

It is to be observed that, Π is the difference of two squares of expressions of the form

$$A\Pi^{1/2} = D\Pi^{1/2},$$

the radical which multiplies the being

$$\frac{1}{a + b\,\cos T + c\,\cos T'\sqrt{ab}} \cdot \sqrt{ab - a_1\cos T - b_1 \cos T'};$$

the differential becomes

$$\frac{dT \cdot \sqrt{ab - a_1\cos T - b_1\cos T'}}{(\cos^2 T \cdot \frac{a+b}{1+\alpha\,\cos T'} + \beta\,\sin^2 T)\,\sqrt{a + b\,\cos T + c\,\cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q\,\cos T + R\,\cos T'}\sqrt{a + b\,\cos T + c\,\cos T'}}(a+b\,\cos T + c\,\cos T')(1+\alpha\,\cos T+\beta\,\cos T').$$

(Page 208.) The formula thus becomes

$$\frac{4}{\Pi}\,\Delta A = a\,a'\Big[M\Big(\frac{\partial U}{\partial x}\,a' - \frac{\partial U}{\partial y}\,a\Big)(\Omega\omega + a)\,2b'\xi - \frac{\partial U}{\partial x}(ba - b'a)(\xi-\xi')\Big]$$

$$-\tfrac{1}{2}\Pi b'(ab - a_1b_1)(c - c_1) - \Pi ab\,(a+b)(\xi - \xi')(\sqrt{XY})$$

$$+\tfrac{1}{2}\Pi\big(abcdc' - (\sqrt{XY})\big) - \omega_1(\xi-\xi')(\xi-\xi').$$

The second formula thus, suppose

$$\Pi = -(a+b+c+d+e+f) - \tfrac{2}{3}(a+b+c+d+e+f)^2 - \tfrac{1}{3}(a+b+c+d+e+f)^3$$

$$-\tfrac{1}{2}\Pi b'\big(abcdc' - (\sqrt{XY})\big),$$

and the value of Π thus, suppose

$$\Pi = \tfrac{1}{2}\,\big(\sqrt{abcd} - \sqrt{abcdc}\big)(a_1 - b_1)\,ab,$$

or so they may be written,

$$\Pi = \tfrac{1}{2}(abc - a_1b_1c_1)(bcd - a_1c_1d_1) + a^2 - 2axy + a^2c_1\sqrt{XY}.$$

The above two expressions involving M have on the right-hand side terms which, when collected with the second of those given in the formulae of §8, give the final expression ΔA, viz.,

$$\Delta A = aa'M\,(\xi-\xi')\,\Pi(bc' - b'c)(bc - b'c') + \cdots$$

which is obviously a sum of squares.

(Page 209.) *Completion of the reduction and final expression for* ΔA.

But now substituting the values of Π, Π', W, we find

$$\frac{2}{\Pi}\,\Delta A = aa'\Big[(aa+ba+a'a+b'a)(\xi-\xi')\,a+(\xi-\xi')ab_1'-(\xi ab_1'a'b_1')\,2c_1\,(\sqrt{XY})$$

$$+(a'a'b_1'-a'b\,a'b_1)(\xi-\xi')a+(a'b_1'-a'b\,a_1'b_1')(\sqrt{XY})$$

$$-(\Pi a'b_1'b_1+aba_1'b\,a_1'c_1)(\Omega-\Omega')\,(bw+ac)(\xi-\xi')\Big],$$

viz., the coefficients belong to the partition which contains the factor aa', viz., the expression for $\frac{4}{\Pi}\Delta A$; the other portion contains the factor

$$(a'+a'a+ba_1b_1')\Pi\cdot(acb'cde'-\sqrt{XY}\,(bw'\cdot c_1)\,a'b_1c_1')\,ab\,a'b\,a'b',$$

where

$$X=b\,(dacdef),\quad \text{etc.}$$

We have then the complete result, viz. this is

$$\frac{2}{\Pi}\Delta A = aa'M\,[(a+b)\,2\Pi-a\Theta a+aba'c_1\xi\,(\Omega-\Omega')\,aaa'\,\Omega-a\,c'a'b'c\,a'b'c\,(\Omega-\Omega')]$$

$$+\cdots$$

which is obviously a sum of squares.

(Page 210.) Also

$$-(a'a'U+U)(ba'U+U_1)=-\Theta b'a'(-aa+aa_1-a'b'+bac)\,U_1\,cb\,b'c'c_1\,U_1=-a+ba,+2aca'a_1';$$

adding these two quantities, the right-hand side divides by (a_1-a), viz., by Θ, the resulting value is

$$=W=b'(aa+a'a'+aa'a'b_1')+a'b_1(aa+ba_1'baa')\,a_1'b_1,$$

we then see that this radical coefficient of $(ba')^2$ is

$$=2\Theta b'(ba'a)\,2c_1W'b'.$$

To this is to be added the actual coefficient of $2\Theta a$ in the next step

$$2\Theta b'(b'a'a)\,2c_1\,W'a',$$

and we see that, without difficulty

$$(\Theta b'(a'a)\,2\Theta b'\,a_1'b)\,W'c\,a'b'c\,a'c'a'b\,aa'a'\,c_1c_1\,W=0.$$

Lastly, for the coefficient of $(\partial\xi')^2$, we have

$$\Big(2\,\frac{\partial U}{\partial x}\,a'-2U(\alpha+a_1)\Big)^2-\Big(2\,\frac{\partial U}{\partial y}\,a'-2U'(\alpha'+a_1)\Big)^2$$

$$=4(\alpha-\alpha_1)(b+b_1)(c+c_1)(d+d_1)(e+e_1)(f+f_1)(b+b_1')-8acc_1'(ba_1b'a+b_1ac)\,(ba'a'b'b'b_1+aca_1'a)$$

$$+8abcd(cc'-ba)(bc-b'c')(b'c-bc')(\sqrt{XY});$$

which is the required value.

(Page 211.) Adding it to the preceding expression, the sum is

$$= (a'-a)(a+b)(a+c)(a+d)(\sqrt{XY})(aa'b_1'+a'b\,a_1'b_1')(a+b)(c+aa_1'+ba_1'a+ac)\,a+a'a'\,c\,c'(ab_1'c_1')$$

$$+]\cdots - 2b'(aa+aa'c)\,a'(aa+a'a'c_1')\,c_1\,a + 2c'\,(aaa+ac')\,a + 2b_1a_1'c_1'\,(\sqrt{XY})$$

$$+\cdots + (2b' - baa' - a'b)\,4a_1'b_1 + 2c_1(\Pi a'ba' + a'c)\,4a'b_1\,c_1.$$

This is, in fact, divisible by $(a_1 - a)$; that is, by Π; for there exists between the symbols the relations

$$ba'b' + a'c' + c\,a'a'c\,a'b'c'a'c'a'a',$$

$$b'cba'a'c\,a'b\,a'a'ca' + a'b\,(c\,a'c'b'a')b'a',$$

$$c'a'b'c'a'(aaa'c_1'c_1'a'c_1'a'c'a_1')$$

$$+(2b' - 2a'aa'c'a'cc) = 0,$$

and we thus reduce the expression to

$$\frac{2}{\Pi}(ba' - b'a)(b + aa'a'b'b'b_1)(cb'b_1b_1 + c'c_1')(c_1' - a'b_1')(\xi - \xi')(c + c')$$

viz. effecting the division, the quotient is

$$= 2b' - 2b'c\,(b+a) - 4a'b'a'c_1'(c_1' - cc_1'a'c_1) - (aaa'c_1')(\sqrt{XY});$$

and we thus reduce the coefficient of $(\partial\xi)^2$ in like manner

$$\Pi_a' = (b'-a)(\sqrt{XY}) - (aa'c_1')(\sqrt{XY});$$

viz., this is Π_a' multiplied by $(b'-a)(ac'-a'c)(b'c-bc')(b-b') = -(b-b')(aa'c_1')(\sqrt{XY})$.

(Page 212.) or finally it is

$$\Pi_a' = -2c'(a'-a)\,(2b+2a'-a'c) + (a'b'-a'c)(b+b'-a) - 2(b+a'-a')(a+a'-a')$$

$$+2(aa'-a'b')(b-b'+2a'c'+a_1'c)\,2b'c'\,(b+cb'a_1')(b_1'+a+2bc'a_1')$$

$$+2a'c' - 2b\,a'c\,a'c + b'a'(aaa_1' + bb'c_1' - a'b_1') + b'a'a'c'c_1' + aca'a_1' + 2a'c_1'$$

$$+(c'-a_1')(c_1'+a'b)(\sqrt{XY}) - 2a(a'c)(ac'\,b_1)(c_1' + b'c_1'a_1'b_1')$$

$$+2bac_1'.$$

It is to be observed that the investigation thus far has been entirely independent of the values of Π, Π'; those values are, in fact, such as to make the coefficients of $\sqrt{XY}$ vanish; for the actual numerical values of these coefficients can then be found, and it may easily be confirmed with each other: the foregoing investigation of those values was therefore unnecessary.

(Page 213.) Hence the case of paper 665 ends with the establishment of the formula

$$A\Delta + A^2\Pi = 0;$$

or, more concisely, $\Delta A = 0$.

Cambridge, 7th December, 1877.

2 [666] Sur un exemple de réduction d'intégrales abéliennes aux fonctions elliptiques

[From the Comptes Rendus de l'Académie des Sciences de Paris*, t.* LXXXV *(juillet–décembre, 1877), pp. 265–268 ; 373, 374 ; 426–429 ; 472–475.]*

Je reprends l'investigation de M. Hermite sur lui rapport aux intégrales réductibles

$$\int \frac{(1,x)\,dx}{\sqrt{a\cdot 1-x\cdot 1+ax\cdot 1+bx\cdot 1-abx}},$$

publiée au-dessus du titre : "Sur un exemple, etc.," (*Annales de la Société scientifique de Bruxelles*, 1876).

Nous avons les constantes a, b et les variables x, y, u, v ; et on posant

$$X = a\cdot 1-x\cdot 1+ax\cdot 1+bx\cdot 1-abx,$$

$$Y = a\cdot 1-y\cdot 1+ay\cdot 1+by\cdot 1-aby,$$

(où $\sqrt{a\cdot 1-x\ldots 1+bx} = X$, etc.,) et les fonctions de l'intégration, les fonctions elliptiques, par fonctions elliptiques, on a les équations différentielles

$$\frac{du}{\sqrt{X}} + \frac{dv}{\sqrt{Y}} = \frac{1}{\sqrt{a}}(du+dv),$$

$$\frac{x\,dx}{\sqrt{X}} + \frac{y\,dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}\sqrt{b}}(du-dv).$$

Il y a en effet trouvé les expressions, en rayon des fonctions elliptiques de u, v, des fonctions symétriques $x+y$, xy, et de, dx, dy, des fonctions a, b, c, d, e, f des fonctions ab', cde, … ou (avec une notation plus simple) ab, ac, ad, … ces fonctions sont d'ailleurs les fonctions de M. Hermite.

(Page 215.) Précisément, ces fonctions sont

$$a = ay, \quad b = 1-x\cdot 1-y, \quad c = ax\cdot 1+ay, \quad d = 1+bx\cdot 1+by, \quad e = 1-abx\cdot 1-aby;$$

$$ab = \sqrt{a(1-x\cdot 1+ax\cdot 1+bx\cdot 1-abx\cdot ay\cdot 1-y\cdot 1+ay\cdot 1+by\cdot 1-aby)} = \sqrt{XY},$$

$$ac = \sqrt{a(1-x\cdot 1+ax\cdot 1+bx\cdot 1-abx\cdot ay\cdot 1+ay\cdot 1+ay)}\cdot(1+bx)(1+by) - aby\cdot(x-y),$$

$$ad = \sqrt{a(1-x\cdot 1+ax\cdot 1-aby\cdot 1+by\cdot 1-aby)} = -1+ax\cdot 1+by\cdot 1-x\cdot 1+by\,(x-y),$$

$$ae = \sqrt{a(1+bx\cdot 1+by\cdot 1-abx\cdot 1+aby)} = +1+bx\cdot 1+by\cdot 1-aby\,(x-y),$$

$$af = \sqrt{a(1-abx\cdot 1-aby)} = -1+abx\cdot 1+aby\cdot(x-y),$$

$$bc = (1-x\cdot 1+ax\cdot 1+bx)\sqrt{ay} - (1+ay)(1-y\cdot 1-aby)\sqrt{ax} = (1+ay)(x-y),$$

$$bd = (1-x\cdot 1+ax\cdot 1+bx\cdot 1-abx)\sqrt{ay} = (1+by)(x-y),$$

$$be = (1-x\cdot 1+bx\cdot 1-abx)\sqrt{ay} = (1+by)(x-y) = (1+by)\,(x-y),$$

$$bf = (1-x\cdot 1-aby) = (1-aby)(x-y),$$

$$cd = (1+ax\cdot 1+by) - (1+bx) - (1+ay) - (1+by) = (1+by)(x-y),$$

$$ce = (1+ax\cdot 1-abx) - (1+by\cdot 1-aby) = (1+by)(x-y),$$

$$cf = (1+ax\cdot 1-abx)\sqrt{1-aby} - (1+ay\cdot 1-aby)\sqrt{1-abx} = (1+bx+ay)(x-y),$$

$$de = (1+bx\cdot 1+by) - (1+ax\cdot 1+ay) = (a-b)(x-y),$$

$$df = (1 + bx \cdot 1 - abx) = (a - b)(x - y),$$

$$ef = (1 + bx \cdot 1 - abx - 1 + by \cdot 1 - aby) = (1 + bx + by)\,(x - y).$$

et je remarque que la différence du carré quelconque des fonctions ab, ac, ... est une fonction rationnelle en entière dans x, y. On a, par exemple,

$$ac - ab = 1 - x \cdot 1 + ax,$$

$$bc - bc = -1 + ay \cdot 1 + ax\,(x - y),$$

$$cd - cd = -1 - aby \cdot 1 + aby\,(x - y),$$

$$ef - ef = a(1 - b)(x - y),$$

etc.

En désirant, comme auparavant, $a = \sqrt{1 + a \cdot 1 + b}$, et puis

$$\alpha = \sqrt{1 - x}, \quad \beta = \sqrt{1 - y}, \quad \gamma = \sqrt{1 + ax}, \quad \delta = \sqrt{1 + ay},$$

$$\epsilon = \sqrt{1 + bx}, \quad \zeta = \sqrt{1 + by}, \quad \eta = \sqrt{1 - abx}, \quad \theta = \sqrt{1 - aby},$$

soit

$$S = \alpha\,\beta + \gamma\,\delta,$$

et puis

$$\sqrt{ab} = \tfrac{1}{S} \cdot \gamma\,\delta - \beta\,\delta,$$

$$\sqrt{ac} = \tfrac{1}{S(\beta - \beta_1)\sqrt{1 - \alpha b)xy}} \cdot \left[(1 + by)\eta\,\theta - (1 + bx)\eta\,\theta\right] + a\,(by - bx)(b - a)\eta\,\theta,$$

$$\sqrt{ad} = \tfrac{1}{S(\alpha\sigma\beta + \delta_1\sigma\beta)\sqrt{1 - abxy}} \cdot \left[(1 + by)\eta - (1 + bx)\eta\right] + a\,(\alpha - 1)(by - bx)\eta\theta,$$

$$\sqrt{ae} = \tfrac{1}{S} \cdot \left[(1 + by)\eta - (1 + bx)\eta\right] + a\,(\alpha + 1)(by - bx)\eta\,\theta,$$

$$\sqrt{af} = \tfrac{1}{S}(-abxy + abab)\eta\,\theta,$$

$$\sqrt{bc} = \tfrac{1}{S} \cdot \left[(1 + by)\eta - (1 + bx)\eta\right] + a(by - bx)\eta\,\theta,$$

$$\sqrt{bd} = \tfrac{1}{S}(a + b)\,(by - bx)\,\eta\,\theta,$$

$$\sqrt{be} = \tfrac{1}{S}\,(ay - ax)\,\eta\,\theta,$$

$$\sqrt{bf} = \tfrac{1}{S}(-1 - aby) \cdot (-1 + abx)\,\eta\,\theta,$$

$$\sqrt{cd} = \tfrac{1}{S} \cdot \beta - \sigma\,\delta_1 a\,(c\,b)\,\eta\theta,$$

$$\sqrt{ce} = \tfrac{\beta - \sigma_1\delta_1}{S}\,a\,(c\,b)\,\eta\,\theta,$$

$$\sqrt{cf} = \tfrac{1}{S}\,(1 - abx \cdot 1 - aby - 1 + ax \cdot 1 + ay)\eta\theta,$$

$$\sqrt{de} = (a - b)\,(x - y),$$

$$\sqrt{df} = \tfrac{1}{S}(a + b)(b - a)\,\eta\,\theta,$$

$$\sqrt{ef} = \tfrac{1}{S}(-ax\,1 - aby + 1 + ay\,1 - abx)\eta\theta.$$

(Page 216.) (où j'écris en, du plus, $\sin am$, $\cos am$, Δam,) et, pour ses moments,

$$\xi = \sqrt{ax\,(1 - y)(1 + ay)} = \tfrac{xy}{\sqrt{a}\,\eta - xy\,\theta},$$

$$\eta = \sqrt{ay\,(1 - x)(1 + ax)} = \tfrac{xy}{\sqrt{a}\,\xi - xy\,\theta},$$

et, qui sont des fonctions elliptiques, je donne l'identification :

$$\xi^2 = (\xi - \xi)(\eta - \eta)\, a\, b = \xi^2 - \eta^2 - a \cdot b.$$

On a l'identité (due à M. Hermite)

$$\big(P\eta + Q\eta + R\eta - S\xi\big) - \big(P\xi + Q\xi + R\xi - S\eta\big)(a\,b\xi + a\,c\xi + a\,d\xi + a\,e\xi + af\xi) = (\xi^2 - \eta^2)(\xi^2 + \eta^2 - ab),$$

où

$$\xi = a \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \quad \text{et} \quad \eta = 0 = \xi = ax.$$

On a ξ identité (due à M. Hermite)

$$(\xi P + \eta P + \xi R + \eta R - \xi S - \eta S)(\xi^2 + \eta^2 - ab) = (\xi^2 - \eta^2)\, \eta\, \xi\, ab.$$

Or, l'identité (due à M. Hermite) :

$$P = \alpha\, \beta + a\, \gamma\, \delta\, (\eta\theta + b\, \eta\theta),$$

$$Q = \alpha\, \beta - \gamma\, \delta\, a\, (\eta\theta + a\, \gamma\theta),$$

$$R = \alpha\, \beta - a\, \beta\, \gamma\, (\alpha\beta + \gamma\delta) - (\eta\theta + a\, \eta\theta),$$

$$S = \alpha\, \beta\, \gamma\, \delta + (\beta\, \beta\, \gamma\delta - \gamma\delta)\, \beta\delta,$$

et je suppose l'équation

$$P + Q + R + S = a\, xy\, (\alpha + \beta + \gamma + \delta) - a^2 c^2.$$

En faisant remplacement $x = b\,b$, $y = a\,a$, on a chaussement reduit la différentielle aux signes des radicaux à une certaine forme particulière, savoir, par les expressions rationnelles des radicaux qui figurent dans les expressions de $\sqrt{ab}$, $\sqrt{ac}$, $\sqrt{ad}$, ...

$$\sqrt{ab} = \tfrac{ab}{1 + a\sqrt{ab}} \cdot \xi\, (1 + b\, b) - \big((1 + a^2) + \alpha\xi^2 \cdot (1 + b)\big)\sqrt{ab}\, \xi\eta.$$

(Page 217.) (Pp. 373–375 ; 426–429.) Les valeurs de $x+y$, xy donnent une beaucoup de point faible de a, b, c, d, e, f, mais les réductions pour obtenir les valeurs de a, b, c, d, e, f des fonctions ab, ac,... ne sont pas présentées, je donne maintenant ses résultats. Ces valeurs sont

$$\sqrt{a} = \tfrac{a}{\sqrt{aS}} \cdot \gamma\, \delta - \beta\, \delta,$$

$$\sqrt{b} = \tfrac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot \big[\delta(1 + by)\sigma\, \eta - (1 + bx)\eta\big] + a\, (\eta - \theta)\, a\, \eta\, \theta,$$

$$\sqrt{c} = \tfrac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot \big[\delta(1 + ay)\, \eta - (1 + ax)\, \eta\big] + a\, (\eta\theta + \alpha b\, \theta)\eta\, \theta,$$

$$\sqrt{d} = \tfrac{1}{\sqrt{(1-\alpha\beta)-\delta_1}}\sigma\eta - \tfrac{1+ay}{\sqrt{aS}}\, \xi\, \eta + \cdots,$$

$$\sqrt{e} = \tfrac{1}{\sqrt{aS}}\, \xi\, \eta + \cdots.$$

où

$$\xi = a \cdot \eta\, ab\, \eta\, b\, \xi.$$

et puis

$$\sqrt{xy} = \tfrac{1}{\sqrt{S}}\, (\gamma\, \delta - \beta\eta'\, \delta),$$

$$\sqrt{xy} = \tfrac{1}{\xi\, (\beta_1\sigma_1 + \delta_1\sigma_1)\sqrt{(1-\alpha b\, xy)}} \cdot \beta\big[(1 + by)\eta - (1 + b\, b\, b)\big]\eta\theta + (1 + by)\, a\, b\, c.$$

(Page 218.) Les valeurs de a, b, c, d, e donnent $\sqrt{X}\sqrt{Y}\sqrt{XY}$:

$$\sqrt{X}\sqrt{Y}\sqrt{XY} = \sqrt{a\,b\cdot a\,c\cdot a\,d\cdot a\,e\,a\,f}$$

$$= \tfrac{a}{\sqrt{a\,S}}\cdot\big[(\eta\,\theta - a\,b\,\eta\theta)(1+ay)(1+ax)(1+ay\,b)\big] + ((\eta\theta + b\,\eta\theta)\sigma\,c\,d\eta\theta).$$

J'ai vérifié que le signe d'antenne avec celui de la valeur d'origine au moyen à la fois de l'expression rationnelle de $\sqrt{X\,Y}$.

On vérifie en partie les valeurs des fonctions $\sqrt{ab}$, $\sqrt{ac}$, … en considérant les différences des carrés de ces fonctions ; mais on calcule n'est pas toujours facile. Par exemple, nous avons

$$ac - ab = (\alpha b)\,(\beta\alpha - \beta\alpha),$$

et ainsi mieux dans deux façons à peu près égales. Par exemple,

$$\big(ac - ab\big)\big(ac + ab\big) = \tfrac{ab}{S}(\sigma\sigma + \eta\eta)\,a^2 - a^2\sigma\eta\theta\,(\eta\theta + b\,\eta\theta),$$

et cette valeur doit être égale aux fonctions $\sqrt{ab}$, $\sqrt{ac}$, … qui ont toujours faciles. Par exemple,

$$ac = \tfrac{a^2}{(bS\beta - bS\delta)}\eta\theta\,b\,\big[(1+by)\eta - (1+bx)\eta\big] + a\,(\eta\theta + b\eta\theta)\,a\,\eta\,\theta.$$

Pour cette fois, j'écris cette expression sous la forme

$$A = \alpha\,\beta + ab\,c\,c\,c\,(\eta\theta + b\,\theta\theta)\eta\theta,$$

où $A = b'(1+ay)(1+b\,ax)$, et c'est de l'expression rationnelle, on a

$$B = \tfrac{1}{4}(\alpha cc - bcc)(\alpha\beta + \beta\delta) - B,$$

où $B = b'(1+ay)(1+ax)$, et c'est de l'expression rationnelle ;

$$V = \beta(cc'')\,a\,d\sqrt{a\,b\,c\,d}\,e\,\xi\,ac\,e\,b\,\eta\theta - 5b\,e\,b\eta\theta.$$

en si ce que c'est l'équation

$$(a + Q + R + S)(\alpha P + \beta Q + \gamma R + \delta S) - A\,B\,(\alpha\beta + \gamma\delta) - A\,B(\alpha\delta + \beta\gamma);$$

c'est-à-dire :

$$P\,V\,e^2\,Q\,V\,R^2\,V^2\,S^2 - A^2(\alpha\beta + \gamma\delta)^2 - B^2(\alpha\delta + \beta\gamma)^2 = 0,$$

et il y a une vérification ainsi complète.

(Page 219.) La différence $bc - ad$ se donne un exemple beaucoup plus simple ; on a

$$bc - ad = 1 + a\,b\cdot 1 - bx\,b\,(x+y),$$

$$= \tfrac{ab}{S}(\beta^2 - \delta^2)\,a\,b\,\eta\theta,$$

$$= \tfrac{a^2b}{S}\,4\alpha\beta\,\beta\delta\,a\,b\,\eta\theta,$$

l'équation à vérifier en ainsi juste.

(Pp. 472–475.) Je donne quelque autre formules dont je me suis servi dans le cours de cette recherche. Partant des expressions de ξ, η, on a

$$d\xi = \tfrac{1}{2}b\,b\,a\sqrt{a\,b}\,b\,\big[(-abx\,b + b\eta - b\theta - 2b^2b\,b)\,dx + (a\,x\,(-abxb + ax - ayb)\,dy\big],$$

$$d\eta = \mu\,b\,a\,b\,c\,b\,e\,\big[(-b\,\theta\,b + b\eta\,(b\,c + b\,a\,c'\,b)\,dx + (b\,(-b\,\theta\,b + b\,b - 2b^2b\,b))\,dy\big],$$

$$d\xi'' = a\,c\,b\,e\,\big[(c\,(-b\,\theta\,b - b\,c\,d - 2b^2\,b\,b))\,dx + (-2b\,(c - a\,c\,b + ab - 2a\,c\,b\,b))\,dy\big];$$

on prenant pour A, B, C des fonctions triples qui

$$A\,d\xi + B\,d\eta + C\,d\xi'' = d\,a,$$

ou

$$A\,b + B\eta + C\,c = 0;$$

on pose

$$A\,b + B\eta + C\,c'' = a\,b,$$

et on obtient de ces équations, j'obtiens pour A, B, C les valeurs

$$\sqrt{a}\,V = \tfrac{1}{\eta\theta}\,(-U + W),$$

$$\sqrt{a}\,V = \tfrac{1}{\xi}\,(U - V),$$

$$\sqrt{a}\,V = \tfrac{1}{\eta}\,(U - W),$$

lesquelles satisfont, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

Réciproquement, on obtient les identités, ce qui est mes pour les peines, en obtient une démonstration de l'équation différentielle

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = \tfrac{1}{\sqrt{a}}(du + dv).$$

(Page 220.) où

$$U = E\,(ba\,c\,e + b\,a\,c\,e) + b\,c\,a\,b\,c\,a\,b\,e \cdot (\Theta\,c + b\,b\,a\,e\,b),$$

$$W = K\,V \cdot (ba\,c\,e + b\,a\,c\,e) + b\,a\,b\,c\,a\,b\,c\,a\,e \cdot (\Theta\,e + b\,c\,c\,b),$$

$$V = E\,V \cdot (b\,a\,c\,e + b\,a\,c\,e) + b\,e\,a\,b\,c\,a\,b\,c\,e \cdot (\Theta\,b + b\,b\,a\,e\,b)$$

$$\nabla = E \cdot (b\,a\,c\,e) + b\,a\,c\,d\,e \cdot (\Theta\,a + b\,b\,a\,e\,b\,c).$$

et de même

$$\Theta\,V = \tfrac{a+b}{2\sqrt{ab}}\,(U - V + W),$$

$$B\,V = \tfrac{a}{2}\,(U - V),$$

$$C\,V = \tfrac{1}{2}\,(U - W).$$

En admettant l'équation

$$\tfrac{dx}{\sqrt{X}} + \tfrac{dy}{\sqrt{Y}} = \tfrac{1}{\sqrt{a}}\,(du + dv),$$

on obtient sur l'extension prise par celles relations

$$AE = \tfrac{x+y}{\sqrt{ab}} = \tfrac{(c''-a)-c}{2}\,V,$$

$$B\,\tfrac{V}{c''-a-b} \cdot V = -a\,b\,b - a\,b - a\,V,$$

$$C\,\tfrac{V}{c''-a-c} \cdot V = -a\,a - b\,a - c\,V,$$

et, en multipliant par

$$b\,V \cdot (c'' - a)\,V = \tfrac{a+b}{2\sqrt{ab}}\,(c'' - a)\,V,$$

et dans les seconds positions, en lieu de

$$c''B\,(b'' - a''), \quad B\,b\,(c'' - a)\,V,$$

substituant les valeurs

$$Px + Qx + Rx + Sx, \quad Py + Qy + Ry + Sy - 1,$$

on obtient, après quelques calculations étranges,

$$e''\Delta\, a = -\, a^2\, V + 2\, c\, a\, V - 2\, a\, b\, c\, V + \alpha c\, (Px + Qx'' + Rx''_1)\sqrt{QV}\, V,$$

$$\nabla\, V = \alpha b(Q\, x + Q\, y'') + \beta\, (Py'' - V + Q\, Q\, V)\, V + V\, C\, V\, Q\, V V,$$

laquelle satisfait, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

(Page 221.) En dérivant, pour plus de simplicité,

$$AV = -\,\tfrac{1}{2}\, U\, V, \quad BV = \tfrac{1}{2}\, UV, \quad CV = -\,\tfrac{1}{2}\, U.$$

les valeurs de AV, BV, CV sont

$$A = \gamma\, \alpha\, V - \gamma\, V,$$

$$B = (-\, \alpha + b\, c\, V)(\Theta\, c + b\, V)\, c\, c\, b\, e + a\, c\, (\Theta\, V - \beta\, V)\, V,$$

$$C = (1 - 2\, a\, c'')(c'' - a + b)\, V + (1 - 2\, acc'')\Theta c\, a\, b\, c\, c\, e + c\, b\, (\Theta\, a + \beta\, V)\Theta V,$$

$$+\tfrac{1}{2}(\alpha b\, c + ab\, c\, b\, e) \cdot V - V + b\, e\, V\, a\, V\, b\, c\, V,$$

et on évite trois équations pour

$$AE,\; B\, \tfrac{C}{\eta\theta},\; C\, \tfrac{C}{a\theta},$$

on dérive

$$\tfrac{V}{\sqrt{ab}}\, XV = R\eta + \cdots,$$

$$\tfrac{V}{\sqrt{ab}}\, bV = a \cdot (\, - \cdots),$$

$$\tfrac{V}{\sqrt{ab}}\, cV = \Omega \cdot (\cdots),$$

où

$$\Omega = \alpha\beta + \gamma\delta, \quad V = \alpha\delta + \beta\gamma.$$

c'est de moyen de ces équations que j'ai trouvé les valeurs ci-dessus données de $\sqrt{ab}$, $\sqrt{ac}$, …, ce qui m'a permis d'aller bien plus loin dans la réduction.

(Page 222.) On a aussi le point à pas de $\sqrt{ac}$ a $\frac{1}{2}\, \Pi\, U\, V$

$$\sqrt{ab} = -\, V + \sqrt{\tfrac{1}{2}\, \Pi\, V},$$

$$\sqrt{ac} = (1 + a)\, V + \sqrt{\tfrac{1}{2}\, \Pi\, V},$$

$$\sqrt{cd} = \tfrac{1}{2}\, (-\sqrt{ab}\sigma V + V\, (1 + a)\, (1 - ab)\, V\, a),$$

$$\sqrt{ce} = \tfrac{1}{2}\, (\xi - \sqrt{ab}\sigma + V\, (1 + a)(1 - ab)\, V),$$

mais il y a des dénominateurs variables qui contiennent des facteurs dont quelques-unes diffèrent des combinaisons, et la réduction aux formes ci-dessus données est traitable à peine de petite.

3 [667] On the Bicircular Quartic. Addition to Professor Casey's Memoir "On a New Form of Tangential Equation."

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Professor Casey communicated to me the MS. of the Memoir referred to, and he has permitted me to make to it the present Addition, containing further developments on the theory of the bicircular quartic.

Starting from his theory of the fourfold generation of the curve, Prof. Casey shows that there exist series of inscribed quadrilaterals $ABCD$ wherein the sides AB, BC, CD, DA pass through the centres of the four circles of inversion respectively; or (as it is convenient to express it) the points of A, B, C, D are belong to one of the same four points of inversion respectively, and they lie or are conditional respectively by depending on the relation between the two parameters. But each such numerical expression $\sin\theta$, θ', θ'', θ''', *say,* or, as respectively, may here put be obtained, are by mere consideration with each other; the foregoing investigation of $ABCD$, and giving to it an infinitesimal variation, we have four infinitesimal arcs AA', BB', CC', DD'; these values are differential expressions AA' and BB' being of the same magnitudes; $M_A da_1$, $M_A db_1$, the identity of the two parameters. But each such numerical expression M_A, M_B, M_C, M_D has not just an analytical form expressed in the formulae of §6, viz.,

$$M_A da_1 = M_B db_1, \quad \text{etc.}$$

where the sum of these expressions is for a, b, c, d unique cancellation in the relation between a, b, c, d which causes the cancellation of this expression $\Delta\Delta E$.

(Page 224.) I propose to complete the analytical theory by establishing the canonical equations $\Delta = R\,a - R\,da$, $\Delta = R\,b - R\,db$, etc., and the relations between the parameters a, a_1, u_1, u_2, which belong to an inscribed quadrilateral $ABCD$.

The curve being the imaginated form is a quadrinodal form

$$\Delta(\,x\,;\,a\,b) \;=\; \Pi\,(a_1 - a\,;\,b_1 - b);$$

wherein each term is symmetrically composed in the differential coefficients of a, a_1, u_1, u_2, and the relations x are conditional respectively to a quadrinomial form

$$\Delta\,a = \Pi(\sin\,a_1\,a + \sin\,a_2\,a)(\sin\,b_1 b + \sin\,b_2\,b)\,du.$$

I remark that in the various formulae f, g, θ, β, f_1, g_1, θ_1, β_1, etc. we can assume the relations $du = 0\,du$.

Where the value of inversion, viz. taking X, T as current coordinates, the equations are:

$$(Px + Qy + Rz)^2 = 1, \quad (X-a)^2 + y^2 = \rho_a^2, \quad \text{etc.}$$

I remark that the analytical change of f, g, θ, β etc. are constants in respect to the elements of the points x, y, and the relation between x, y, u, u_1 is consequence

$$\frac{X}{f-g} + \frac{Y}{g-h} = 1, \quad (X-a)^2 + y^2 = \rho_a^2, \quad \text{etc.},$$

so that x, y, $\theta_x = \frac{a\cos\,a + b\,\sin\,a}{2}$ are functions of a single parameter a, and similarly (u_1, y_1), (u_2, y_2), (u_3, y_3) are functions of the parameters a_1, a_2, a_3 respectively.

Formulae for the fourfold generation of the Bicircular Quartic. Art. Nos. 1 to 5.

1. We have four systems of a designed conic and circle of inversion, each giving rise to the bicircular quartic as the envelope of a generating circle resting in a designed conic, having its centre on the conic, and orthogonal to the circle of inversion. The conic in the second generating circle. Taking X, Y, f, g values, suppose

$$\frac{X^2}{f+g}+\frac{Y^2}{f+h}=1, \quad (X-a)^2+Y^2=R_2^2,$$

the new values, $X=u+Y$, and $Y=R\,b\,a$, give in like manner

$$\frac{X^2}{f+g+g_1}+\frac{Y^2}{f+h+g_1}=1, \quad (X-a)^2+Y^2=R_2^2,$$

$\sqrt{YX}=R_2$, and so for the other two generations.

2. The eight systems are not, however, all four different, but the bicircular quartic represented by $f+g_1$, $g+h_1$, g_1+h, g_1+h_1 etc., and the like circles of inversion, give the same set of curves and, as the figure of the centre of the curve, may be expressed in the form $X=(\xi+\eta)f+(\xi+\eta)g_1+(\xi+\eta)g_1$ (which are, in fact, identically true unless an addition for some other reason).

3. The notations being thus altered, we make to general remarks that the variable x, y, may be considered as the related quartic, taking x, y, as before, that are the equations of the second curve

$$(X^2+Y^2-h)(X^2+Y^2-g)+(g+h)(X^2+Y^2)-2aX-2aa+a^2-g\,h=0,$$

and to obtain the more, a, f, g, h, ρ, are introduced into these expressions, which are connected respectively as R_1, R_2, etc.

$$(X^2+Y^2+f)(X^2+Y^2+g)+(X^2+Y^2+h)\rho^2+(R^2-gh)(g+h)f+(\rho^2gh)=0,$$

which is, in fact, derivable from the bicircular quartic by writing therein X, Y in place of x, y.

The expressions of $(Xx)^2=(x-X)(x+Y)+Y$, where the values may also be written

$$P=2bc\,(u+a)=b_1\,c\,c\,b_1+b_1\,c\,c\,b_1,$$

$$P_1=2bc\,(u_1-1)(b_1-c)=b_1\,c\,(b_1\,c-c_1\,c)(b\,c)(b_1\,c).$$

(Page 225.) Here

$$\sqrt{f+g}, \quad \sqrt{(f+g)(f+g_1)}, \quad \dots$$

indicate the bicircular quartic and the circle of inversion mentioned in the foregoing remarks; in particular, the centres are confined,

$$\sqrt{f+g}\cdot X=(f+a)\,P\,a+(Y-P\,b)\,a^2,$$

$$\sqrt{f+h}\cdot Y=(Y-P\,b)\,a+P\,ba^2,$$

or also

$$(X-a)^2+Y^2=R^2;$$

and we have, by the relation

$$(f+g_1)\cdot(f+h_1)=(a^2-R)^2.$$

(Page 226.) **2.** The eight systems are not, however, all four different, but the bicircular quartic represented by f, g_1, h_1, and f_1, g, h_1, viz., in particular, that the centres are co-linear with one another, may be expressed in the form

$$X=(a+b\,c\,\sin\,a)\,f+(a+b\,c\,\sin\,b)\,g+(a+b\,c\,\sin\,c)\,h,$$

which are, in fact, identically true unless an addition for some other reason.

4. The eight systems are, however, all of them four different. Considering a given series of inscribed quadrilaterals, having its centres on the four conics of inversion respectively, the cancellation in the equation we have $a_1 a_2 = \rho^2$ (where there exists between the curves expressed in the relations of the same conic and circle of inversion).

If b may be remarked that a circle of a pre-imaginary radius u, containing a series of inscribed quadrilaterals $ABCD$, having its centres on the four conics of inversion, of the same radii respectively can be regarded as a determined curve, called *Imaginary circle, etc.*; we shall not have, however, in either of pure imaginary radius.

Investigation of aa'.

5. The coordinates of a point on the designed conic

$$\frac{X^2}{f+g} + \frac{Y^2}{f+h} = 1$$

may be taken to be $(f+g)\,x = (g+g)$, and we hence pass as follows the fundamental

$$\sqrt{f+g+g}\,x = \left[\left(2+a\,a'\right)\left(\xi^2+\eta^2\right) - \left(2(\xi^2-\eta^2)+(\xi^2+1)\right)\right]\xi^2, \quad \text{etc.}$$

(Page 227.) the matter is of the bicircular quartic. Consider the generating circle, centre $(f+a, g+a, h+a)$, which cuts at right angles the circle of inversion

$$(X-a)^2 + Y^2 = R_2^2.$$

If for a moment the variable is called θ, then the equation of the generating circle is

$$(X-a)^2 + Y^2 - R_2^2 = \cos\theta\, a + \sin\theta\, b,$$

the condition for the equation of the generating circle to be inscribed in right angles to the circle of inversion gives

$$(X-a)\xi + Y\eta = (\xi+a)\xi + (\eta+a)\eta = (\xi+a)\xi + (\eta+a)\eta - 1.$$

If for a moment the variable is called θ, then the equation of the generating circle is

$$(X-a)\xi + Y\eta = \cos\theta + \sin\theta + a.$$

But from the equation for X we have

$$B = (X-a)(X-g+Y)^2 - R'(2a-g+g)\,a,$$

and hence between X, Y as the equation of the generating circle, we have

$$(X-a)X + (Y-a)Y = \tfrac{1}{2}(\xi-\eta-2g)X - R'(2a-g+a)\,X.$$

This is, in fact, an inscribed quartic.

6. If we make further to notice that, if ξ denote the inclination to the axis of ξ of the tangent on the designed conic, then the central, viz., the ratio of $f+g+g$ to $g+g$ is the same as η, viz. Casey's θ, then

$$\xi = \frac{\sec\xi}{\sqrt{X}}, \quad \eta = \frac{\sec\xi}{\sqrt{Y}}, \quad \text{where} \quad U = (f+h)\sin^2\xi + (g+h)\cos^2\xi - h;$$

viz., $\sec^2\xi = U$.

If now further be noticed that, if a denote the inclination to the axis of the conic, then the central angle of a related point of point will have

$$a = \frac{\sin\xi}{\sqrt{U}}, \quad y = \frac{\cos\xi}{\sqrt{U}}, \quad \text{where} \quad U = (f+h)\sin^2\xi + (g+h)\cos^2\xi - h;$$

giving, as is easily verified,

$$\frac{du}{u} = \frac{dy}{x}; \quad \text{we have therefore}$$

$$\frac{du}{u} = \frac{dy}{x}, \quad \frac{du}{X} = \frac{(-a+y\,x)}{\sqrt{U}};$$

or

$$\frac{(u+g)^2}{(g+g)^2\xi^2 + (h+g)\,x^2} = du.$$

(Page 228.) or say the two values are

$$B = \tfrac{1-a\xi-\beta y+g\xi}{\xi^2+y^2}, \quad B' = \tfrac{1-a\xi-\beta y-g\xi}{\xi^2+y^2};$$

to preserve the generality it is necessary to introduce a determinate

$$dX = R\,da + a\,dR,$$

$dY = R\,da + g\,dR.$

But from the equation for X we have

$$B = (X-a)(bX)y - 2by + R^2(2a-X)f - X(2R\,dX) = 0,$$

whence

$$dX = R\,da\,(Xx + Yy)\,da,$$

$dY = R\,da\,(Rda - Xfy).$

12. The differentials du, dy can be expressed in terms of a single differential du, viz. writing

$$\xi = \frac{\sec\xi}{\sqrt{U}}, \quad y = \frac{\cos\xi}{\sqrt{U}}, \quad \text{where} \quad U = (f+h)\sin^2\xi + (g+h)\cos^2\xi - h;$$

$$\Omega = (f+g)\,\xi^2 + (f+h)\,y^2,$$

then we have

$$du = \frac{a\xi + by}{\sqrt{a^2+b^2-g-h}}\,da, \quad dy = \frac{-ay+b\xi}{\sqrt{a^2+b^2}}\,da.$$

It is to be observed that, since the designed conic the circle of inversion intersect in elliptic, as a real angle. The designed circle, however, must be of two squares of expressions of the form

$$(f+g)^2 - 2(af+g)(f+g)\,da^2, \quad (f+h)^2 - 2(af+h)(f+h)\,da^2;$$

where $\sqrt{(f+g)(f+h)} = b\,a$ is the distance of the centres of the conic and circle of inversion; and in $\sqrt{Pf}$ is, of course, of the sign of multiplication.

(Page 229.) drawn in the senses A to B, B to C, C to D, D to A: and representing the inclinations, measured from the positive direction of axis of ξ, of these tangents by ν_1, ν_2, ν_3, ν_4 respectively; then, in passing to the consecutive quadrilateral, a, b, c, d change, suppose, into $a+da$, $b+db$, $c+dc$, $d+dd$; and the points A, B, C, D into A', B', C', D', such as $a+da$, $b+db$, $c+dc$, $d+dd$; and we have

$$dA = R\,da, \quad dB = R\,db,$$

$$dC = R\,dc, \quad dD = R\,dd \cdot da,$$

and thence

$$dA = bda - bdA, \quad d_1 da + db_1\, da,$$
$$dB = b\,da - b\,dB,$$
$$dC = b\,dc - b\,dC,$$
$$dD = b\,dd - b\,dD,$$

which are the correct signs in regard to the particular figure.

Reduction of $\int \frac{1\,da}{(\xi^2+y^2)\sqrt{U}}$ *to Elliptic Integrals. Art. No. 28.*

29. The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})\,b - B\,Y - \alpha)^2}}{(c+g)\,\xi^2 + (h+g)\,y^2 + \alpha + a^2 \sin^2 \xi}\,(\cos T + b)^2 \cdot c^2 \sin^2 \xi;$$

where $\sqrt{U}$ is a more constant; and we may apply it to the Gaussian transformation,

$$\cos a = \frac{a + b\,\cos T + a^2 \sin T}{c + a \cos T + b\,\sin T},$$

$$\sin a = \frac{c + b\,\cos T + b^2 \sin T}{c + a \cos T + b\,\sin T},$$

where the coefficients a, b, c, a', b', c', a'', b'', c'' are such that identically

$$\cos^2\, a + \sin^2\, a = 1 = (a + b\,\cos T + a\,\sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2;$$

(Page 230.) so that, $b = \frac{\sin a}{\sqrt{c}}$, $y = \frac{\cos a}{\sqrt{c}}$ are functions of a single parameter a, and similarly (u_1, y_1), (u_2, y_2), (u_3, y_3) are functions of the parameters a_1, a_2, a_3 respectively.

Examples of $\Delta\Delta = 0$: *numeral evaluation. Art. Nos. 14 to 22.*

14. Writing the formulae thus

$$AP + BQ + CR + DS = 0, \quad (a^2 - g^2)\,(f + g + g)\,a^2\,P + a^2\,Q + a^2\,R + a^2 D'^2 + ac\,S^2 = 0,$$

which leads to many discoveries by means of a single quadrinomial. The case where the central b has been added so that the bicircular quartic was expressed by the form

$$\frac{X^2}{f + g + g_1} + \frac{Y^2}{f + h + g_1} = 1, \quad (X - a)^2 + Y^2 = R_1^2,$$

which leads to many discoveries by means of a single quadrinomial. We have, in general

$$\Delta = \tfrac{a+b}{2ac}\,(\Pi a + \Pi b + \Pi c) - 5ac\,(b + cb),$$

and similarly

$$AC + BD = (a-b)(b-c)(b+c)\,AB + (b-a)(c-b)(b+c)\,CD,$$

which is the value of the generating circle.

15. We have

$$H = b_1 - c_1 = \frac{ac_2 + b}{(b-a)(b-c)} \cdot \cos h_1 - \tfrac{ac}{(b-a)(b-c)} \cdot \sin h_1,$$

$M_a = aR'^2 b$, $aR = M_b$, etc., as the lower may be seen,

$$H = \tfrac{a+a}{(b-a)(b-c)} \cdot \cos h_1 - \tfrac{ac}{(b-a)(b-c)}\,(b+a)\,c.$$

We thus have the second expression which has equal absolute value as A;

$$v_1 a' = v_2 b', \quad v_1 b' = v_2 a',$$

of a', b', the expressions

$$P_1 = aacdb - b'cd(ba^2 - cba),$$

$$P_2 = c'a'b'(ba^2 - cba),$$

$$P_3 = R\,b'(a-b)(b-c)\,a + b\,Rc'\,(b-a)(c-b),$$

$P_4 = a^2cb - cb'a$, $P_5 = acb - cba$, $P_6 = cba - abcba$.

(Page 231.) Adding it to the preceding expression, the sum is

$$(a-b)\sqrt{a^2+b}\,a{+}b\,c\,a{-}b\,ac{+}b{-}cb\,c\,a{+}a\,b{-}c\,a\,d{+}a{+}a\,b\,c\,a{+}a\,c\,b\,c{+}bac{+}b\,a\,c{+}a\,b\,c{+}bc\,b\,(ba)^2{+}a^2,$$

$$+]\cdots - 2b'(aa+aa'c)\,a'(aa+a'a'c_1')\,c_1\,a + 2c'\,(aaa+ac')\,a + 2b_1a_1'c_1'\,(\sqrt{XY})$$
$$+\cdots+(2b' - baa' - a'b)\,4a_1'b_1 + 2c_1(\Pi a'ba' + a'c)\,4a'b_1\,c_1.$$

This is, in fact, divisible by $(a_1 - b_1)\,(b_1 - c_1)\,(c_1 - a_1)$, which the case being of an inquiry between the symbols

$$ba'bacccbacacbccacbacdcbacdacbacbaccbac\,(\sqrt{XY})$$

gives the development.

We have therefore

$$\sqrt{ab}\,A = acb'ab' - acb' - b'a - 2acb' \cdot acb';$$

viz., we are dividing the expression by A, we have the result expressed which is reducible by the equality

$$B = -axy(\sqrt{ab}) - (a+a)\,acb'(a+b) = 0.$$

We thus reduce the expression to

$$B = -\,axy\,ac\,(ba)\,ab\,c + ac\,ac\,c\,ab\,c\,a + (\sqrt{ab}),$$

and there seems to be no advantage in further reducing the integral.

20. The relation between (x,y), (x_1,y_1) may be expressed also in the two forms

$$1 - a(\xi+a) - B(\eta+a)\,a + (\xi+a)(\eta-1)\,a = \frac{a+b}{(a+b)(b+c)}\,a + (-a+b)(a+1) - B(-a+a) = 0,$$

$$1 - a(\xi+a) - B(\eta-a) - 2\tfrac{a+b}{(a+b)(b+c)}\,a + (a+b)(a+1) - B(a+a) = 0.$$

In fact, the first of these equations is

$$\Big[(1-aa+B)\,a + (\xi+a)(\eta-1)\,a\Big] - \Big[(a-1-aa+B)\,a + (\xi+a)(\eta-1)\,a\Big] - 2a^2\,a + B^2\,a + b\,2\,ac = 0,$$

which, by reason of the values of a, b, becomes

$$(1+ax+ay)(a+b\,c) - (a+b)\,ac\,a + (-a+b-a)\,a + B\,(a-a) = 0,$$

$$-(ax + ay) - ay - ay + aay\,(a + a) = 0,$$

or that, $b = \frac{ac}{b-a} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} \cdot \sin h_1$.

And in like manner the second equation may be verified.

21. The two equations are:

$$1 - a(\xi + a) - B(\eta + a) = \frac{(a+b)}{(a+b)(a+c)} \cdot \cos\theta_1 - \tfrac{ac}{(a+b)(a+c)} \cdot \sin\theta_1,$$

$1 - ax - ay$, $(b + c)$, $(b - a)\,a + R\,b - R\,c\,(b + 1) = ay\,(a + b)$, which is the resulting value.

In substituting the E^1 and B their values, these are

$$\sqrt{X}\sqrt{Y} = (ax)^2 + (g + a) + (g + h)\sqrt{XY} - ay\,x\,(g + a) + (a + a)(g + h),$$

$X = ay$, $Y = ay$, and so $\sqrt{YX} = \frac{ay\,c}{g} \cdot \sin\theta$ also; and the resulting value, and similarly

$$\sqrt{XX}\,Y = \tfrac{1}{2}(ax + ay)(ax + 1)(ax + 1) - 2ay\,ac\,(g + a)(b + h),$$

also verified by the data.

(Page 232.) and similarly,

$$\sqrt{X}\sqrt{Y}\sqrt{XY} - (ay + ay)(b + g)\sqrt{XY} - ach^2,$$

which yields the required expressions.

22. It may be remarked that, if $aa' = 1$, then $X = 1$, $X' = 1$, $Y = 1$, $Y' = 1$, and consequently we have

$$\sqrt{XX'}/(ch\,c\,h)\,(c\,c\,c)\,(h\,h) - XX'(-2g\,h) - 2axy = R = R',$$

and we have these relations

$$(\xi - \xi')(c\,c\,c)\,c\,c\,c = 0, \quad \text{etc.},$$

and these relations

$$\Omega_2 = (\xi - \xi'')ch \cdot c\,c\,c = 0, \quad \Omega_2' = 0,$$

$$(\xi' - \xi)c\,c\,c\,c\,c\,c\,c = 0,$$

and similarly,

$$A\xi + B\eta + C\,c = 0;$$

that is, $\xi = R\frac{a\,c\,c}{c\,c\,c}\,(c\,c\,c) - 2gh$, $\eta = ay$, $\xi'' = ay$, and so for the others.

The expressions of $(\xi x)^2$ etc. are, however, very simple. Two of them are,

$$W = \sqrt{XX'} + \sqrt{YY'} - 2\sqrt{ay}\sqrt{ay\,b - c}(ay + axb) - g\,(c - c)a^2 - 2agxy,$$

$$W = \sqrt{XX'} - \sqrt{YY'} - 2\sqrt{ay}\sqrt{ay\,b - c}(ay + axb) - g\,(c - c)a^2 - 2agxy.$$

The other two values $(\xi y)^2$ may be similarly written.

(Page 233.) **20.** The relation between (x, y), (x_1, y_1) may be expressed also in the two forms

$$1 - a(x + x_1) - B(y + y_1)\,a + (x + x_1)(y - 1) - a = (a + b)(b + c)\,a + aac - c(a + 1) - B(a + a) = 0,$$

$$1 - a(x + x_1) - B(y - y_1) - 2\tfrac{a+b}{(a+b)(b+c)}\,a + (a + b)(a + 1) - B(a + a) = 0.$$

In fact, the first of these equations is

$$\big[(1 - aa + B)\,a + (\xi + a)(\eta - 1)\,a\big] - \big[(a - 1 - aa + B)\,a + (\xi + a)(\eta - 1)\,a\big] - 2a^2\,a + B^2\,a + b\,2\,ac = 0,$$

which by reason of the values of a, b, becomes

$$(1+ax+ay)(a+b\,c)-(a+b)\,ac\,a+(-a+b-a)\,a+B(a-a)=0,$$

$$-(ax+ay)-ay-ay+aay\,(a+a)=0;$$

or, substituting for E^1 and B, their values, these are

$$\sqrt{X}\sqrt{Y}=(ay)^2+(g+a)+(g+h)\sqrt{XY}-axy\,(g+h);$$

$X=ay$, $Y=ay$ and so $\sqrt{YX}=\frac{ay}{c}\,c\,c\,c$ also.

21. The two equations are

$$1-a(x+x_1)-B(y+y_1)=\frac{a+b}{(a+b)(a+c)}\cos\theta_1-\frac{ac}{(a+b)(a+c)}\sin\theta_1,$$

$$1-ax-ay,(b+c),(b-a)\,a+R\,b-R\,c\,(b+1)=ay(a+b),$$

which is the resulting value.

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(Page 234.) **22.** As a partial verification, I remark that $\Delta\Delta$ should be symmetrical in regard to the constants b, c, d, e, f; this is obviously the case as regards b and f, and as in regard to the two in h; if we interchange b and c this coefficient of b' becomes

$$=w_1(ab+b+c+e+d+a)(a+d+e+f),$$

$$-(-2a+b+c+a+a+a+d+d)\,c'\,(b\,c)\,a+b\,cd+(b\,c\,d+bac)\,(c\,c\,c+e\,d)\,c\,c\,ac\,a\,(\sqrt{XY}),$$

and if we interchange the relations b between this coefficient should be in the formed

$$-=w_1(ab-b-c-e-d-a)(a+d+e+f),$$

$$-(-2a-b-c-d-d-a)\,c'\,(b\,c\,a)+bcd\,(b'\,ac+aac\,a\,b)\,(cc+dc)\,a+(-2a-c)\,a\,a\,(\sqrt{XY}),$$

$$=2\Delta\Delta;$$

which should be the case.

It is to be observed that the investigation thus far has been entirely independent of the values of R, R', R'': those values are, in fact, such as to make the coefficients of $\sqrt{X\,Y}$ vanish; for the actual numerical values of these coefficients can then be found, and it may be easily confirmed with each other; the foregoing investigation of these values was therefore unnecessary.

Completion of the reduction and final expression for $\Delta\Delta$.

But now substituting the values of R, R', R'', we find

$$\tfrac{2}{R}\,\Delta A=aa'M\,[2c'\big(bc-b'c\big)+2(b+c)(c+b)]-2bc\,(bc-b'c')+2a'b'(-a+b_1'+c)$$

$$-a^2(c+b)(c-b')-a^2(b+b')(\xi+\eta')(-ab'+c)$$

$$+\tfrac{1}{2}a'b'c'(aa'-ba)(\sqrt{XY})-b'a'c_1'c(\sqrt{XY}),$$

$$-2(c+b)c\,a+bc_1\,a_1\,a_1-2aa'(c-c_1')\,a'(c-c_1')\,a',$$

$$-(-2a+b+d-a+a)\,c'\,(b+c)\,c'(b+a)\,dc'\,a+c\,d-bc\,a\,b\,a+ad-ba+(b-c)-2c_1\,b\,(ba+c\,a)c_1'$$

$$-4c'\,c\,(c-ac\,ac)(bc-b\,c)\,a\,c+4ac'\,c'(ac'-ac)c\,(a'-a)\,b'\,c'c_1'\,ac_1'\,(\sqrt{XY}),$$

which is obviously a sum of squares.

(Page 235.) in terms of the relation: just found between ξ and $a\,a'$, these two sets of values will agree together if only

$$R'\,(Y-(g+h)\,a)=a(Y-X)\,a,$$

$$R'\,(X-(h+a)\,a)=R\,(Y-X)\,aaa.$$

These are easily verified: the first of them, viz.,

$$RY-R'\,(g+h)\,a=a(Y-X)\,a,$$

and the second values give, in like manner,

$$ax\cdot\frac{ac(c-ac)}{a}(ac_1'-1)(bc-ac)\,c\,c'(a'-a)\,a',$$

where, on x each ξ, η, ξ', η', ξ and a values, with the equation

$$R\,(R^2-2aa')(b-X)\,(a+aa)\,(a'-a)=0;$$

this proved by means of the relation

$$(R^2-2aa')(b-X)\,(a+aa)\,(a'-a)\;=0,$$

where $a=R\,a$, $b=R\,c$, $c=R\,c$, ..., prove the identity of the two relations of the value $a'-a$, prove the identity of the two relations of the values $a\Delta a'$.

Considering the quadrilateral $ABCD$, then da is the element of arc AA', viz. if ν is the inclination to the axis of ξ of the tangent at A drawn in the direction A to B, then da, da are the projections of AA' on the axes of ξ, η respectively, and so for the other elements b, c, d. And similarly the foregoing the of values are

$$\Pi\,A\,B=R\,b\,a\,(R^2-2aa'),$$

$$\Pi\,C\,D=R\,b\,c\,(R^2-2aa'),$$

$$\Pi\,B\,C=R\,c\,d\,(R^2-2aa').$$

Formulae for the Elements of Arc a, b, c, d, Πa, Πb, Πc, Πd, etc. Art. No. 23 to 27.

23. The formulae are

$$da=-vR\,c\,\sqrt{X}\sqrt{Y}\,\frac{ds}{\sqrt{X}\sqrt{Y}}=\nu\,R\,c\,\frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$db=-vR\,c\,\sqrt{X}\sqrt{Y}\,\frac{ds}{\sqrt{X}\sqrt{Y}}=v_1\,R\,c\,\frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dc=-vR\,c\,\sqrt{X}\sqrt{Y}\,\frac{ds}{\sqrt{X}\sqrt{Y}}=v_2\,R\,c\,\frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dd=-\nu R\,c\,\sqrt{X}\sqrt{Y}\,\frac{ds}{\sqrt{X}\sqrt{Y}}=v_3\,R\,c\,\frac{ds}{\sqrt{X}\sqrt{Y}},$$

where the ν each denote ± 1. Supposing as above that ν is to be changed into ν', ν'', ν''' are positive: then R, R have opposite signs: but R, R have the same sign.

(Page 236.) as here also R' and R_1, and R_1' and R_2. We may take R, R_1, R_2, and R_3 all of them positive: the value of A_1, viz.,

$$\Pi_a\,a=a\,M_a\,b-ba\,M_b\,a=aacb\cdot da,\quad\ldots$$

hence to make dA, dB, dC, dD all positive, we must have M_a, M_b, M_c, M_d each of them of the same sign: for which ν_1, ν_2, ν_3, ν_4 must reciprocally,

$$dA = v_1 R\, b \frac{ds}{\sqrt{X}\sqrt{Y}} = aaM_b \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dB = v_2 R\, c \frac{ds}{\sqrt{X}\sqrt{Y}} = aaM_c \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dC = v_3 R\, d \frac{ds}{\sqrt{X}\sqrt{Y}} = ac\, M_b \frac{ds}{\sqrt{X}\sqrt{Y}},$$

$$dD = v_4 R\, a \frac{ds}{\sqrt{X}\sqrt{Y}} = ad\, M_c \frac{ds}{\sqrt{X}\sqrt{Y}};$$

must hence follow dA, dB, dC, dD in either case in the form positive: $a, b, c, d, \nu_1, \nu_2, \nu_3, \nu_4$ all to be 1 or 0, all to be either both positive or both negative respectively.

24. But from $R = -2\,aa$, $b = 2aa'$, …, we have, in either case in the form positive

$$\frac{1}{(a'+y')\sqrt{X}\,\sqrt{Y}} = \frac{1}{(a'+y')\sqrt{X}\sqrt{Y}} = \frac{1}{(a'+y')\sqrt{X}\sqrt{Y}} = P, \quad P_1, \quad P_2, \quad P_3,$$

$$dS = dS_1 = 2\,aa'(b'a),$$

$$ds_1 = 2acP_1,$$

$$dS_2 = 2acP_2,$$

$$ds_3 = 2acP_3,$$

and consequently

$$dS = \nu R\,(ac\,P + ac\,P_1) - \alpha c(a + ac\,P_1)\,a + ac\,P_2,$$

$$dS_1 = \nu R\,(ac\,P + ac\,P_1) - \alpha c(a + ac\,P_1)\,a + ac\,P_2,$$

$$dS_2 = \nu R\,(ac\,P + ac\,P_1) - \alpha c(a + ac\,P_1)\,a + ac\,P_2,$$

$$dS_3 = \nu R\,(ac\,P + ac\,P_1) - \alpha c(a + ac\,P_1)\,a + ac\,P_2,$$

which are the required formulae for the elements of arc.

25. The determination of the signs has been made by means of the particular figure: but it is easy to see that the points of terms could not for instance be all of them A, B, C at the interior corner of the curve, and at the exterior corner; or we could not have A, B, C, D all four positive, D at the exterior of the curve, A at the interior, and the like a similar examination would not be possible to eliminate dA, dB, dC, dD, and them obtains an equation such as

$$b'\,dA + 2a'\,bacP + a'c\,bacP_1 = 0;$$

(Page 237.) this would, by virtue of the relations between da, db, dc, dd, become

$$\tfrac{Y\,b\,a}{(ax)^2+Y\,a^2}\,dx + \tfrac{X\,b\,a}{(ax)^2+Y\,a^2}\,dy + \tfrac{X\,b\,a}{(ax)^2+Y\,a^2}\,dz + \tfrac{Y\,b\,a}{(ax)^2+Y\,a^2}\,dw = 0,$$

an equation between the variables in abc, … which cannot have a quadrinomial expression admitted by the variable form.

26. The differentials da, dy may be observed, the foregoing the of values $\Pi\,dx$

$$= \tfrac{\xi\,dx}{(a+g)(b+h)}\,X + \tfrac{Y\,dy}{(a+g)(b+h)}\,Y,$$

where $\sqrt{ab} = \sqrt{f + g\,a + b\,g}$. It is also observed by the form θ', and

$$b'' = b + b'\xi\xi'\xi'\xi - 2(b - h - \xi\xi - g\,a + a)\,a + aa'(c - c)\,\xi_1',$$

viz. in like manner be calculated in θ as a function of a and a, an irrational function $M\,\xi - \xi - X\sqrt{U}$, and it would be the rest of such a function. Considering a diagonal conic and the corresponding circle of inversion, the corner of the curve, called Π, we have however, in elider:

$$\frac{d\,\Theta}{da} = \frac{f - a}{\Pi'\,\Pi a + f\,a + a},$$

and the formula thus becomes

$$2\,V\,ab\sqrt{ab}\,da = \frac{ab}{\Pi'}\,(\Pi a + fab\,a)\,da,$$

consequently

$$\Theta\,a = -(\sqrt{ab} - \sqrt{ab\,c})\tfrac{1}{2}\cdot a + ac'\,(\sqrt{ab\,c})(ac'),$$

where k is the rather of the squares of the differentials; and these conditions and (a, b, c, d, e, f) are the are conditional respectively a determinate sign ν.

27. The system of curves it

$$ds = aa\,ac\,da + ac\,db + ac\,b\,dc + ac\,b\,da + ac\,da,$$

$$ds_1 = aa\,ac\,ba + ac\,b\,dc + ac\,b\,b\,dc + ac\,b\,da + ac\,b\,da,$$

$$ds_2 = aa\,ac\,dc + ac\,b\,c\,da + ac\,b\,da + ac\,db + ac\,da,$$

$$ds_3 = aa\,ac\,dc + ac\,b\,dc + ac\,b\,b\,dc + ac\,b\,da + ac\,b\,da,$$

where $\nu_a = \pm 1$, $\nu_b = \pm 1$, … for, etc., are conveniently used in writing it.

Reference to Figure. Art. No. 28.

28. I constructed a bicircular quartic consisting of an exterior and interior oval with the following numerical data: f, g, h, ρ, g_1, h_1, ρ_1, g_2, h_2, ρ_2, g_3, h_3, ρ_3, a, a_1, a_2, a_3, … etc. very convenient values, inasmuch as

(Page 238.) the exterior oval cuts (and crosses each, large; the annexed figure shows 0, 1, 2, 3, the centres of the circles of inversion, the interior oval, and a portion of the exterior oval, the centres 1, 2 outside the exterior oval; I add further the values

$$\sqrt{f + h_2}\,a = 95, \quad \sqrt{-(g + h_2)} = 244, \quad a_2 = 1019, \quad R_2 = -96,$$

$$\sqrt{f + h_1}\,a = 746, \quad \sqrt{g + h} = -241, \quad a = 873, \quad R_1 = 294,$$

$$\sqrt{f + h_1}\,a = 715, \quad \sqrt{g + h} = 245, \quad a = 873, \quad R_2 = -96,$$

$$\sqrt{f + h}\,a = 1934, \quad \sqrt{g + h} = -149, \quad a = 919, \quad R_3 = 23.$$

We thus see how there exists a series of quadrilaterals $ABCD$, where A, B are situate on the interior oval, C, D on the exterior oval. Considering the sides as

(Page 239.) drawn in the senses A to B, B to C, C to D, D to A: and representing the inclinations, measured from the positive direction of axis of ξ, of these tangents by ν_1, ν_2, ν_3, ν_4 respectively; then, in passing to the consecutive quadrilateral, a, b, c, d change, suppose, into $a + da$, $b + db$, $c + dc$, $d + dd$; and the points A, B, C, D into A', B', C', D', such as $a + da$, $b + db$, $c + dc$, $d + dd$; and we have

$$dA = R\,da, \quad dB = R\,db,$$

$$dC = R_2\,dc, \quad dD = R_3\,dd,$$

and thence

$$dA = bda - bdA, \quad d_1 da + db_1\,da,$$
$$dB = b\,da - b\,dB,$$
$$dC = b\,dc - b\,dC,$$
$$dD = b\,dd - b\,dD,$$

which are the correct signs in regard to the particular figure.

Reduction of $\int \frac{1\,da}{(\xi^2+y^2)\sqrt{U}}$ *to Elliptic Integrals. Art. No. 28.*

29. The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})\,b - B\,Y - \alpha)^2}}{(c+g)\,\xi^2 + (h+g)\,y^2 + \alpha + a^2\sin^2\xi}\,(\cos T + b)^2 \cdot c^2 \sin^2\xi;$$

where $\sqrt{U}$ is a more constant; and we may apply it to the Gaussian transformation

$$\cos a = \frac{a + b\,\cos T + a^2 \sin T}{c + a\cos T + b\,\sin T},$$

$$\sin a = \frac{c + b\,\cos T + b^2 \sin T}{c + a\cos T + b\,\sin T},$$

where the coefficients a, b, c, a', b', c', a'', b'', c'' are such that identically

$$\cos^2\,a + \sin^2\,a = 1 = (a + b\,\cos T + a\,\sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2.$$

(Page 240.) Hence

$$(\cos\theta + \sqrt{c-b})^2 + (\sin\theta + \sqrt{c-b})^2 = a^2 - b^2,$$

also

$$\Big[\frac{f-a}{\cos\theta + \sqrt{c-b}}\Big] = \frac{a-b}{f-h}\,(\cos\theta + \sqrt{c-b})\,a^2;$$

$\Pi' = \sqrt{aa' - \sin^2\theta\,(c-b)}$, so that the integral becomes

$$\frac{-a}{\sqrt{(c+g)(c-b)\,c'}}\,dT,$$

which is, as it should be, a function of an elliptic integral.

Cambridge, 7th December, 1877.

4 [668] On Compound Combinations

[From the Proceedings of the Lit. Phil. Soc. Manchester*, t.* XVI *(1877), pp. 113, 114;* Memoirs, ib., *SER. III., t. VI. (1879), pp. 99, 100.]*

Prof. Clifford's paper, "On the Types of Compound Statement involving Four Classes," [volume of *Proceedings* quoted, pp. 88–110 ; *Mathematical Papers*, pp. 1–13], relates mathematically to a question of compound combinations ; and it is worth while to consider its connexion with another question of compound combinations, the application of which is a very different one.

Starting with four symbols A, B, C, D, we have sixteen combinations of the form $ABCD$, ($1{\cdot}4{\cdot}6{\cdot}4{\cdot}1 = 16$), as before). But in Prof. Clifford's question 1 means ΛWCD, Λ means ΛHCD ; viz. each of the symbols means an aggregate of four assertions; and the 16 symbols are then all of them combinations of such assertions; and the question to be considered is to determine the number of types of the binary, ternary, ... combinations of the sixteen combinations; for, according as these are

$$\text{No. of types} = \frac{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}{1, 4, 6, 13, 19, 27, 37, 47, 51, 50, 47, 37, 27, 19, 6, 4}$$

together. In the first mentioned point of view the like question arises, in regard to the new belonging to the five different forms of compound combinations; and as it is worth while to consider its connexion with another question of compound combinations, the formal so the following development.

$$\text{No. of types} = \frac{1, 2, 3, 4, \ldots, 24, 25, 26, 27}{1, 2, 5, 11, \ldots, 11, 5, 2, 1}.$$

(Page 244.) as is very easily verified ; but if the number of letters A, B, ... be greater (say this $= 6$), or, instead of letters, writing the numbers 1, 2, 3, 4, 5, 6, 7, 8, then the question is that of the number of the types of combination of the 20 thirds 13, 14, ..., 78 taken 1, 2, ..., 27 together, a question presenting itself in geometry in regard to the bitangents of a quartic curve (see Salmon's *Higher Plane Curves*, Ed. 2 (1873), pp. 222, et seq.); the numbers, so far as they have been obtained, are

$$\text{No. of types} = \frac{1, 2, 3, 4, \ldots, 24, 25, 26, 27}{1, 2, 5, 11, \ldots, 11, 5, 2, 1}.$$

It might be interesting to complete the series, and, more generally, to determine the number of the types of combination of the $\frac{1}{2}n\,(n-1)$ duads of n letters.

5 [669] On a Problem of Arrangements

[From the Proceedings of the Royal Society of Edinburgh*, t.* IX*. (1878), pp. 338–342.]*

It is a well-known problem to find for n letters the number of the arrangements in which no letter occupies its original place; and the solution of it is given by the following general theorem : viz., the number of the arrangements which satisfy ay, r conditions is

$$\big(1 - \Sigma(1) + \Sigma(2) - \ldots\, (-1)^r\,\Sigma(r)\big)(1 \cdot 2\, \ldots\, n),$$

$$= 1 - \Sigma(1) + \Sigma(2) \cdots \ldots \pm (12\, \ldots r);$$

where 1 denotes the whole number of arrangements; (1) the number of those which fail in regard to the first condition ; (2) the number which fail in regard to the second condition ; (12)

the number which fail in regard to the first and the second conditions; and so on, the second condition; the Σ refers to the combinations of the n conditions which can occur there; and lastly $(1 \cdot 2 \cdots r)$ means $1 \cdot 2 \cdot 3 \cdots r$.

Then, in the special problem, the first condition is that the letter in the first place shall not be α; the second, the letter in the second place shall not be b; and so on. Hence

$$\Sigma_0 = \Pi n - \frac{n}{1}\Pi(n-1) + \frac{n \cdot n-1}{1 \cdot 2}\Pi(n-2) + \ldots \pm \frac{n \cdot n-1 \cdot 1}{1 \cdot 2 \cdots n}\Pi 0,$$

$$= 1 \cdot 2 \cdots n\Big[1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \frac{1}{1 \cdot 2 \cdot 3} + \cdots \pm \frac{1}{1 \cdot 2 \cdots n}\Big],$$

giving for the several cases

$$n = 1, 2, 3, 4, 5, 6, 7, \ldots,$$

$$N_0 = 1, 2, 9, 44, 265, 1854, \ldots$$

I proceed to consider the following problem, suggested to me by Professor Tait, in connexion with his theory of knots: to find the number of the arrangements of n letters $abc \ldots k$, in such manner as to fix the distances of the elements of all those non-pairs aa, bb, cc, $\ldots$ shall be the value k, where k is the value of inversion or expression. This contains the foregoing problem, as the present one is that in which $k = 1$.

(Page 246.) Numbering the conditions $1, 2, 3, \ldots, n$, according to the places to which they relate, a single condition is called [1]; two conditions are called [2] or [1 1], according as the numbers are consecutive or non-consecutive; three conditions are called [3], [2 1], [1 1 1], according as the numbers are consecutive and one not consecutive, or all non-consecutive; and so on: the numbers which refer to the conditions being always written in their natural order, and it being understood that they follow each other cyclically, so that $n = 6$, the set 125 of conditions is [3], as consisting of 3 consecutive conditions; and similarly 1346 in [2 2].

Consider a single condition [1], say this is 1; the arrangements which fail in regard to this condition are those which contain aa in the first place or b; whichever there are thus $2(n-1)$ failing arrangements in any form whatever; and thus

$$2\Pi(n-1) = (n-1)\Pi(n-1).$$

Next for two conditions; these may be [2], say the conditions are 1 and 2; or else [1 1], say they are 1 and 3. In the former case, the arrangements which fail are those which contain ab, or ba, in the first and second places; there are thus, in the first place a, or b, and in the third place c or d; the letters in those places are a, c, a, d, b, c, b, d, in the third place c or d; the letters in those places are a, c, a, d, b, c, b, d; and so on, the number of failing arrangements is thus $= 2 \cdot \Pi(n-2)$. And so, in general, when the conditions are $[\alpha, \beta, \gamma, \ldots]$, the number of failing arrangements is

$$= (\alpha+1)(\beta+1)(\gamma+1)\ldots\Pi(n-\alpha-\beta-\gamma-\ldots).$$

But for $[a]$, that is, for the entire system of the n conditions, the number of failing arrangements is (not as by the rule it should be $= n+1 \cdot bn) = 2$. The number of failing arrangements in any form whatever; for the n conditions abc, abd, $\ldots$ in the conditions [2, 2], [1, 2], etc., would give ξa.

Changing now the notation so that [1], [2], [1 1], [3], &c., shall denote the *number* of the conditions [1], [2], [1 1], [3], &c., respectively, it is easy to see the form of the general result. If, for greater clearness, we write $n = 6$, we have

$$1, \quad -\Sigma(1), \quad +\Sigma(12), \quad -\Sigma(123),$$

$$N_0 = 720 - \{[1] = 6\}\, 2\,|120 + \{[2] = 6\}\, 3\,|\, 24 - \{[3] = 6\}\, 4\,|\, 6$$

$$+\{[1\,1]=9\}\,2\,2+\{[2\,1]=12\}\,3\,2\cdot 2,$$
$$+\Sigma(1234),\quad -\Sigma(12345),\quad +\Sigma(123456)$$
$$+\{[4]=6\}\,5\,|2-\{[5]=6\}\,6\,|\,1+\{[6]=1\}\,2$$
$$+\{[3\,1]=6\}\,4\,2+\{[3\,2]=6\}\,4\,3$$
$$+\{[2\,2]=6\}\,3\cdot 3+\{[1\,1]=6\}\,3\cdot 3.$$

(Page 247.) or, reducing into numbers, this is

$$N_0=720-1296+1296-504+90-36+2=80.$$

The formula for the next succeeding case, $n=7$, may be obtained in like manner

$$N_0=5040-10080+7080-3040+1560-4960+2=579.$$

Then for the preceding cases, $n=4$, 5, 6, respectively we have

$$N_0=4,\quad 24-40+16=0,\quad 120-240+184-60+6+12-2=20,$$

$$N_0=1,2,0,3,80,579,\ldots$$

We have here $r=0$, 4, 6, ... which is, $r=0$, 5, 7, ... respectively $*$.

The formula for general n, considering the second sums above-mentioned, becomes

$$N_0=\Pi n-720+1296-720+(n-1)(n-2)\Pi(n-3)-36+\ldots=0,$$

The formula for the preceding cases of an arrangement a, b, ... may here be put in the form

$$N_0=24-40+16=0,\quad \ldots$$

in respect of the result number of the symbols a, b, c, ... are of n in cyclical order beginning with a, ... $*$a, the letter in the last place not b or c, ...

For the actual variation of so r is $=0$, and the foregoing formulae become

$$n=4,\ 5,\ 6,\ 7,\ \ldots$$

which the analysis a sum of squares.

(Page 248.) all the substitutions which serve every letter. Thus when $n=5$, we obtain the 44 substitutions for these 4 letters b, c, d, e ; these are

(abcde, bdec, cebd, 24 symbols obtained by permuting in every way the four letters b, c, d, e at one time of n

$$(ab)(cd),\ (ac)(bd),\ (ad)(bc),$$
$$(abcd),\ (abdc),$$
$$(acbd),\ (acdb),$$
$$(adbc),\ (adcb).$$

Here in the first column, performing on the symbol $(abcd)$ the substitutions $(abcd)$, we obtain ab ; $(cdcb)$, ab ; *i.e.* to perform a substitution which is again opening with a ; we must in the

0. In explanation of the notation of substitutions, denote that (abcd) means that all the changes may be made in any way 1, 2, 3, ..., but only at the same time; for instance, a is to be changed for b, this b is to be changed for c, this c is to be changed for d, this d is to be changed for a, ... The set (1 abcd) thus all of them denote the same substitution ; the (a 1 d) is to be changed for the cycle (12).

manner end, the substitutions are nothing else than the substitutions which form the column system in the same way. The 13 substitutions are thus of these 4 different types, or any of the symbols which they belong, viz.,

$$adcde, \quad bdcae, \quad cdeab, \quad ddeba,$$

$$ddeab, \quad ebcad,$$

are of 5 different types. The question is to determine for any value of n, the number of the different types, in this case there appear, in this case, but it is not at present yet to find it.

6 [670] [Note on Mr. Muir's Solution of a "Problem of Arrangement."]

[From the Proceedings of the Royal Society of Edinburgh*, t. IX. (1878), pp. 388–391.]*

The investigation may be carried further: writing for shortness w_1, u, v, &c. in place of Ψ (1), Ψ (2), &c., the equations are

$$u_0 = 1,$$

$$u_1 = 0u_0,$$

$$u_2 = 0u_1 + 1,$$

$$u_3 = 0u_2 + 11u_1,$$

$$u_4 = 0u_3 + 6u_2,$$

$$u_5 = 0u_4 + 10u_3 + 12u_2 + 1.$$

Hence assuming

$$u = u_0 + u_1x + u_2x^2 + u_3x^3 + \cdots$$

we have

$$\begin{aligned} u = \frac{1}{1-x^2} &+ u_0\,(2x + 6x^2 + 12x^3 + u_2x^4 + \cdots) \\ &+u_1\,(2x + 6x^2 + 10x^3 + 12x^4 + \cdots) \\ &+u_2\,(4x + 10x^2 + 18x^3 + 28x^4 + \cdots) \\ &+u_3\,(2x^2 + 6x^3 + 12x^4 + 20x^5 + \cdots); \end{aligned}$$

so that, forming the equation

$$\begin{aligned} u'\tfrac{x^2}{(1-x)^2} &= u_0\,(x^2 + 2x^3 + 3x^4 + \cdots) \\ &+u_1\,(2x^2 + 6x^3 + 10x^4 + 12x^5 + \cdots) \\ &+u_2\,(4x^2 + 10x^3 + 18x^4 + 28x^5 + \cdots) \\ &+u_3\,(2x^3 + 6x^4 + 12x^5 + 20x^6 + \cdots), \end{aligned}$$

(Page 250.) where u' denotes $\frac{du}{dx}$, we have

$$u - u\,\tfrac{x^2}{1-x^2} = \tfrac{1}{1-x^2}\,Q,$$

where

$$Q = u_0\left(-\tfrac{x^2}{1-x} + u_1\,2x + u_2\,4x + u_3\,2x^2 + \cdots\right)$$

$$= \tfrac{1}{1-x^2}\left[1 + \frac{2x}{1-x} + \frac{4x}{1-x} + \frac{2x^2}{1-x^2}\right] Q\,(1-\;a\,x)$$
$$= \frac{2x}{(1-x)^2} + Q\Big[1 - \frac{2x}{1-x^2}\Big]\,Q\,(1-\;a\,x).$$

so that the equation is simplified: write

$$u = \frac{1}{1-x}\,Q - \Big(-\tfrac{x^2}{1-x^2} + \tfrac{x^2}{1-x^2}\Big)\,Q,$$

and the equation becomes

$$u'\,\tfrac{x^2}{(1-x)^2}\,Q\;=\tfrac{1}{1-x^2}\,Q + \tfrac{2x}{1-x^2}\,Q\;=\tfrac{1-x}{1+x},$$

then

$$\Big(1-\tfrac{2}{x^2}\Big)\,Q\;=\;Q\,\big(1-\tfrac{2}{x^2}\big),$$

or

$$-\tfrac{(1-x)x^2 - x^2(1-x)^2}{x^2}\;=\;Q\,\Big(1-\tfrac{2}{x^2}\Big)\;=\;1-\tfrac{x}{1+x},$$

or finally

$$Q\big(1-\tfrac{x}{1+x}\big) = Q\,\tfrac{x^2}{(1+x)^2},$$

giving

$$Q = e^{-x}\Big[\int \tfrac{x^2}{(1+x)^2}\,e^x\,dx\Big],$$

and thence

$$u = \frac{1-x}{x^2} + Q\big(\tfrac{1}{x^2} + \tfrac{1}{(1+x)^2}\big)\Big[\int\;\tfrac{x^2 e^x\,dx}{(1+x)^2}\Big],$$

which is the value of the generating function, $u = u_0 + u_1x + u_2x^2 + u_3x^3 + \cdots$.